GAME DESIGN DOCUMENT (GDD): DAHDI

Document Version: 1.0 (Production Master)
Project: Dahdi — Traditional Indian Mills Board Game
Institution: Academic / College Game Development Project
Date: September 2026

PART 1: CONCEPT DOCUMENT

Title Page

- Game Name: Dahdi: Traditional Indian Mills Board Game
- Game Logo: Dahdi title presented in the web interface; no separate logo asset is required.
- Game Catch Phrase: "Place wisely, form the mill, capture strategically."
- Document Type: Full Game Design Document (GDD) & Technical Specification
- Document Version: 1.0 (Gold Master Specification)

Credit Page

- Document Purpose: Authoritative game design and technical blueprint for the digital adaptation of the traditional Indian mills-style strategy game.
- Working Title: Project Dahdi Core
- Game Concept: Digital preservation and gamification of a traditional alignment-and-capture board game, implemented as a browser-based strategy experience.
- Author / Developer: Soumya Ranjan Samal
- Academic Project: College Game Development Project

Sign-Off

Role

Role	Name	Status	Signature Date
Student Developer	Soumya Ranjan Samal	Prepared	September 2026
Project Supervisor	Faculty / Project Supervisor	Pending	—
Project Evaluator	Faculty / Department	Pending	—

Introduction

Dahdi is a digital adaptation of a traditional Indian mills/alignment-style board game. Players place and move nine pieces on a 24-point board arranged as three concentric squares. The central tactical objective is to create lines of three pieces, called mills, and use those formations to capture eligible opposing pieces.

- Genre: Casual Digital Board Game / Traditional Strategy Game.
- Player Type: Casual players, students, families and strategy-game learners.
- Technical Form: Browser-based 2D interface using HTML5, CSS3 and JavaScript modules.
- Theme: Traditional Indian-inspired board-game presentation.
- Design Intentions: Preserve the strategic character of a mills game while making the rules understandable and convenient to play digitally.

Game Analysis

Game Description

Dahdi is a turn-based two-player strategy board game. Player 1 is represented as Peacock and Player 2 as Haldi. In Player vs AI mode, Haldi is controlled by the computer. The board contains 24 playable positions arranged across three concentric squares. Players alternate turns and interact directly with the board points while the rules engine maintains the authoritative match state.

Genre & Elements

- Genre: Traditional Board / Strategy / Alignment / Casual.
- Core Game Elements: 24 board points, nine pieces per side, mills, captures, movement, flying and win conditions.
- Game Content: Turn-based planning, tactical blocking, mill formation and piece reduction.
- Modes: Player vs AI and 2-Player Pass & Play.

Theme & Style

- Theme: Traditional Indian-inspired board-game heritage.
- Style: Clean 2D digital board presentation with clear tokens and legal-move indicators.
- Game Sequence: Placement → Movement → Flying, with capture decisions after mills.
- Player Count: Two sides; one can be AI-controlled.

Game Reference & Taxonomy

- Taxonomy: Deterministic turn-based strategy and alignment game.
- Strategic Immersion: Players evaluate immediate mills, threats, blocking positions and mobility.
- Learning Immersion: The rules are introduced through visible phases, indicators and the Rules screen.

Game Technical

- Technical Form: 2D top-down browser board with DOM/SVG/CSS based visual presentation.
- Camera View: Fixed full-board view so all 24 playable positions remain visible.
- Target Platform: Modern desktop and mobile web browsers.
- Technology: HTML5, CSS3 and JavaScript ES modules.
- Audio: Web Audio API.
- Input: Mouse/touch interaction and keyboard shortcuts.
- Storage: Browser localStorage for selected preferences.

Game Sales & Distribution

- Consumer Group: Students, casual strategy players, families and users interested in traditional games.
- Distribution: Browser-based academic demonstration and local web deployment.
- Monetization: Not part of the academic implementation; the project is designed as a free educational game.

Game Atmosphere

The atmosphere is intentionally calm and board-game focused. The interface combines a traditional-inspired board with modern controls, keeping attention on strategic decisions rather than fast action. Optional Tanpura ambience and soft action sounds provide an Indian-inspired acoustic identity.

Gameplay Walkthrough

1. Match Initialization

Start the game and choose Player vs AI or 2-Player Pass & Play. If AI mode is selected, choose Easy, Medium or Master. The optional diagonal board configuration can also be enabled from Settings.

2. Placement Phase

Each player has nine pieces. Players alternate placing one available piece on an empty board point. A newly completed line of three pieces forms a mill.

3. Capture

After a mill is formed, the player selects an eligible opponent piece for removal. Protected pieces inside mills are normally preserved while other removable pieces exist.

4. Movement Phase

After all pieces are placed, players move a selected piece along a legal connected board path to an empty point.

5. Flying Phase

When a player has exactly three pieces remaining, the Flying phase permits that player to move a piece to any empty point.

6. Winning Condition

The match ends when the opponent has been reduced to fewer than three pieces or has no legal movement available.

Key Features & Selling Features

- Dual Play Modes: Player vs AI and local Pass & Play.
- Three AI Levels: Easy, Medium and Master.
- Strategic Utilities: Undo, Hint, Restart and Rematch.
- Accessibility: Colour-blind mode and reduced-motion option.
- Localization: English, Hindi and Odia interface text.
- Audio: Optional SFX and Tanpura ambience.

Player Definition

Player 1 — Peacock / Side A

The Peacock side is the first player identity used by the game. It is represented through the player card and token set.

Player 2 — Haldi / Side B

The Haldi side is the second player identity. In Player vs AI mode, the computer controls this side.

Pieces

- Nine pieces are available to each player at the beginning of a match.
- Pieces have board positions, ownership and legal movement determined by the rules engine.

Rules and Gameplay

- Players alternate turns.
- Placement uses empty board points.
- A mill is a defined line containing three pieces of the same side.
- A successful mill creates a capture opportunity.
- Protected mill pieces are handled according to the implemented capture rule.
- Movement follows the board's legal adjacency connections.
- Flying becomes available at exactly three remaining pieces.
- The game checks both piece count and legal mobility for the end condition.

Player Rewards

- Mill formation creates an immediate capture opportunity.
- Legal-move indicators provide immediate visual feedback.
- Hint can suggest a useful move.
- Winning triggers celebratory confetti and result feedback.

User Interface (UI)

- Main 24-point board.
- Player status cards and piece information.
- Current turn and phase indicator.
- Legal destination markers.
- Undo and Hint buttons.
- Restart / Rematch controls.
- Settings modal with mode, AI difficulty, language, board variant, audio and accessibility controls.
- Rules/help screen explaining Placement, Movement and Flying.

Heads Up Display (HUD)

- Current turn.
- Current phase: Placement, Movement or Flying.
- Player-side information.
- Selected piece and valid destinations.
- Capture and win feedback.

Player View

The player sees the complete 2D board from a fixed top-down view. There is no scrolling character camera. Keeping the whole board visible supports strategic comparison of mills, threats and movement options.

Antagonistic Elements

The opponent is the main antagonistic element. In single-player mode the opponent is controlled by the local AI; in Pass & Play it is another human using the same device. There are no physical enemies, combat systems or environmental hazards.

Artificial Intelligence (AI)

- Easy: Shallow search with some random choices for a casual challenge.
- Medium: Depth-3 search with move ordering for a stronger tactical challenge.
- Master: Deeper search with iterative/deeper analysis and transposition caching.
- Evaluation: Considers material, mills, threats and mobility.
- Rule Safety: AI uses the same legal-move and rules logic as the human player.

Global Game Elements

- 24 playable points.
- Standard mill configuration plus optional diagonal variant.
- Three gameplay phases.
- Two player sides: Peacock and Haldi.
- Persistent language, audio and accessibility preferences.
- Rules, settings and result overlays.

Story and Theme

Dahdi is not a narrative adventure. Its identity comes from traditional board-game heritage and strategic play. The digital presentation aims to make the traditional rules accessible while preserving the focus on planning, alignment and tactical capture.

Level / Match Design

Dahdi is match-based rather than level-based. Each match contains Placement, Movement and Flying phases. The optional diagonal configuration changes board connectivity and possible mills without requiring separate story levels.

Audio & Sound F/X

- Web Audio API based interface and gameplay feedback.
- Soft UI tap and action sounds.
- Optional Indian Tanpura drone for ambient music.
- Separate SFX and music volume controls.
- Mute and preference persistence through browser localStorage.

Game Architecture

File / Module	Purpose
index.html	Main interface, rules/help text, settings and controls.
constants.js	Board points, players, phases, mills, translations and game constants.
rules.js	MillState / MillRules, board state, legal moves, mills, captures, protected pieces and win checks.
ai.js	Easy/Medium/Master search, evaluation, move ordering and transposition cache.
render.js	Board drawing, token rendering, legal move indicators and win confetti.
audio.js	Audio context, SFX, music and volume/mute management.
app.js	Application controller, state transitions, player interaction, settings, undo, hint and rematch.
CSS files	Main UI, board, tokens and design variables.

How to Play

- Start the game and choose Player vs AI or 2-Player Pass & Play.
- If using AI mode, choose Easy, Medium or Master.
- Place your nine pieces on empty board intersections.
- Try to align three pieces to form a mill.
- After a mill, capture an eligible opponent piece.
- After placement, move pieces along legal board connections.
- When you have exactly three pieces, use Flying to move to any empty point.
- Continue forming mills, capturing pieces and blocking the opponent.
- Win by reducing the opponent below three pieces or leaving no legal move.

Technical Document

- Technology: HTML5, CSS3 and JavaScript ES modules.
- Graphics: DOM/SVG/CSS based 2D board and tokens.
- Audio: Web Audio API.
- Input: Mouse/touch interaction and keyboard shortcuts.
- Storage: Browser localStorage.
- Network: No online multiplayer service required.
- External Engine: No game engine; browser-based implementation.

System Requirements

- Modern Chrome, Edge, Firefox or Safari browser.
- Desktop or mobile device capable of running modern JavaScript and CSS.
- Audio-capable device for sound features.
- Project files served/opened in a browser environment that permits JavaScript modules.

Programming Content

- State-driven game flow.
- Modular JavaScript game state and rules logic.
- 24-point adjacency and mill definitions.
- Legal move generation.
- Mill and protected-piece detection.
- Win-condition checking.
- Negamax / alpha-beta AI search.
- Move ordering and transposition caching.
- Event-driven UI controls.
- Web Audio API based sound management.

Resources

- HTML/CSS/JavaScript development environment.
- Modern web browser for testing.
- Dahdi project source files.
- Optional image/visual editing tools for future board assets.

Project Summary

Dahdi combines a compact traditional strategy rule set with a modular browser implementation. The final design separates rules, AI, rendering, audio and application control so that the project remains understandable, testable and extendable.

Final Design Intent

The completed design prioritizes rule clarity, strategic depth, accessibility and ease of demonstration. The result is intended to function as both a playable academic project and a digital presentation of a traditional Indian board-game style.